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Clinical: Urinary Incontinence

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Innovate.
Reinvent the Future.

Session Objectives

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- Define the different classifications of urinary incontinence
- Articulate the different causes of urinary incontinence
- Note the physiology of normal urinary flow and the pathophysiology of urinary incontinence
- Connect the different treatment options with different classifications of urinary incontinence
- Characterize medical/surgical emergencies presenting as new onset incontinence

Source: Course Syllabus

ScholarRX Bricks

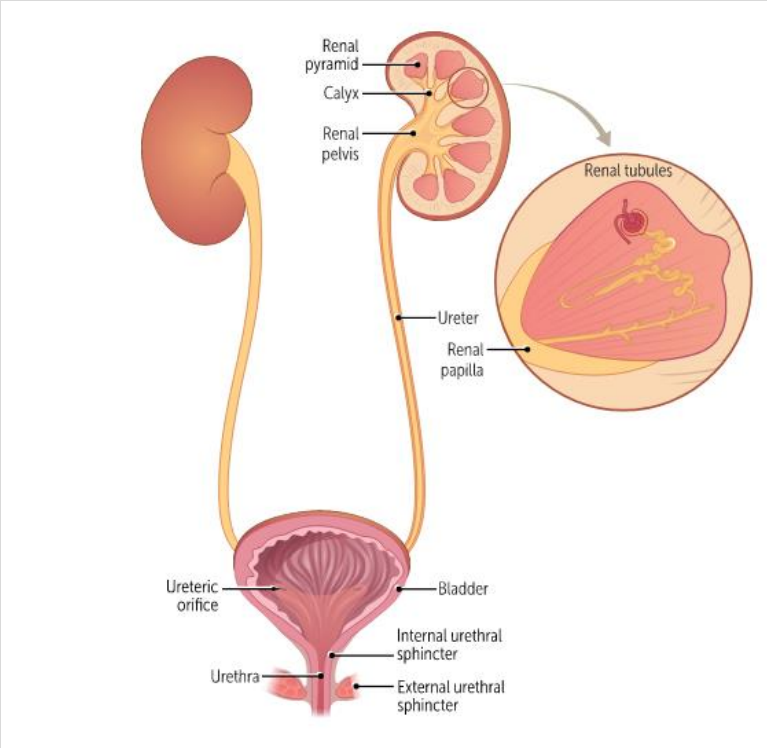
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Micturition and Incontinence

Micturition

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A complex interaction among the bladder, the urethra, the pelvic floor muscles, the endopelvic fascia, and the nervous system maintains continence."



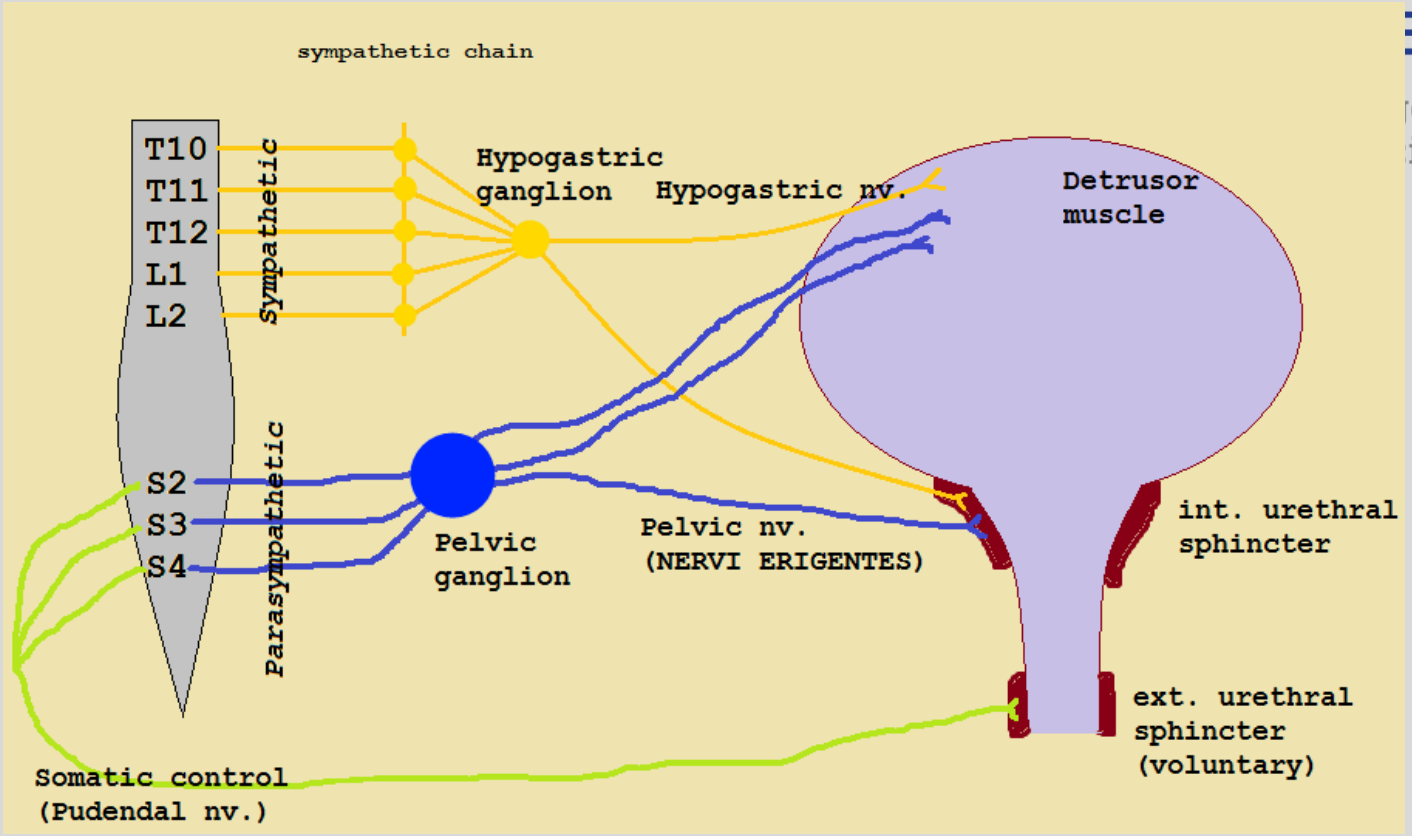
Micturition

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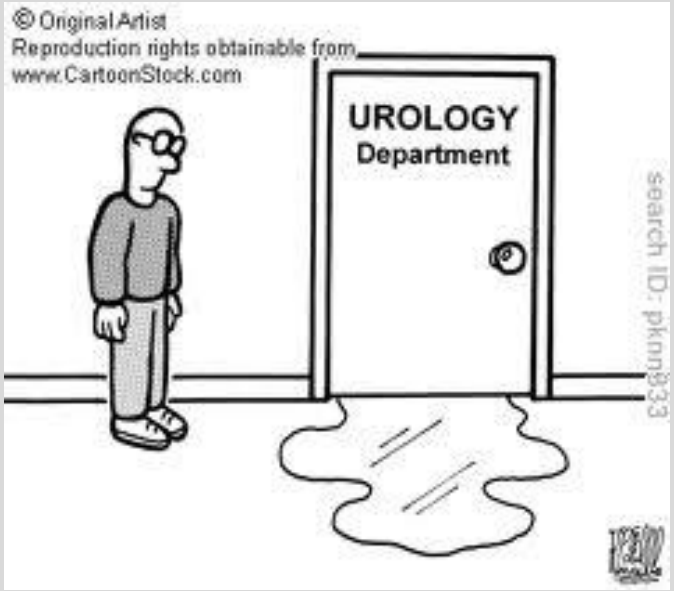
- Micturition requires coordination of several physiological processes
- Afferent: Somatic and autonomic nerves carry bladder volume input to the spinal cord
- Efferent: Motor output innervating the detrusor, sphincter, and bladder
- Cerebral cortex: Predominantly inhibition
- Brainstem- Coordinates urethral sphincter relaxation and detrusor muscle contraction.

Micturition: Pathways

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Incontinence



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Incontinence- involuntary loss of…

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- Stress- …urine with exertion
- Urge- …urine leaking at time of or right after feeling of urgency
- Mixed- …urine at time of exertion with feeling of urgency
- Functional- …inability to hold urine in due to reasons other than neuro-urologic or lower urinary tract dysfunction

Incontinence- involuntary loss of…

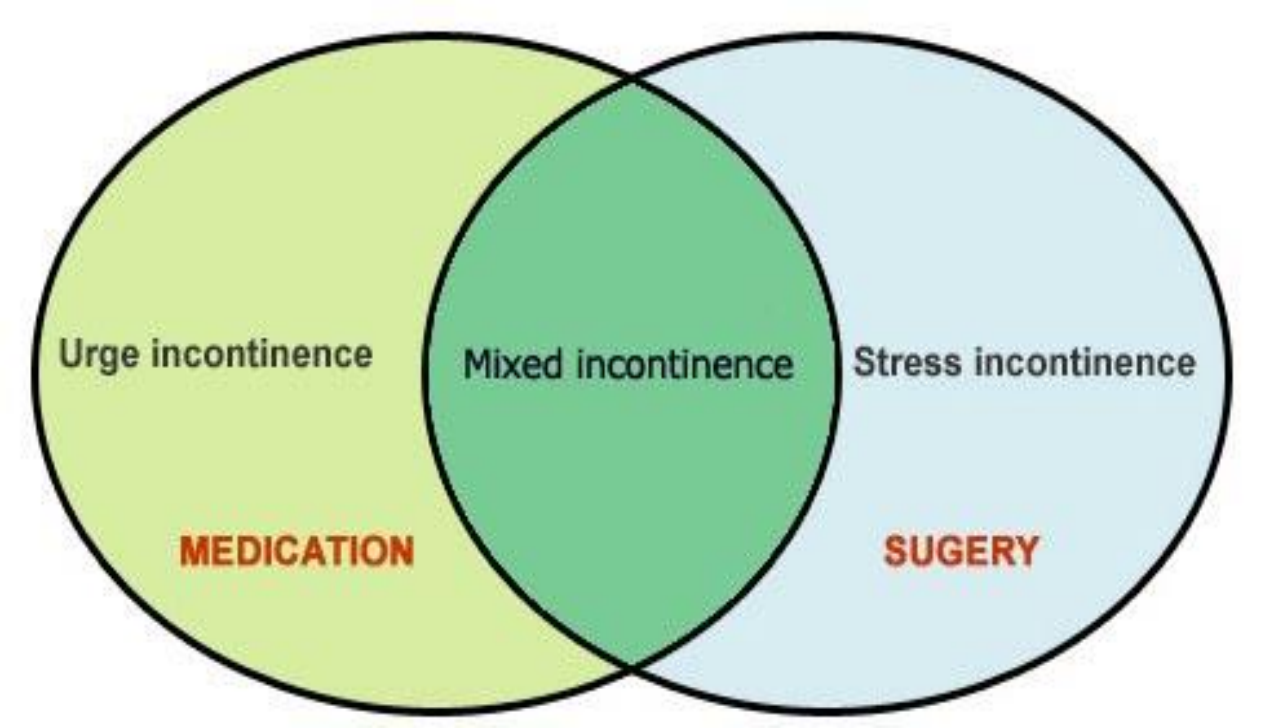
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- Nocturnal enuresis- … urine during sleep
- Continuous- …urine at any time
- Overflow/overactive- Sensation of fullness with incomplete voiding

Incontinence

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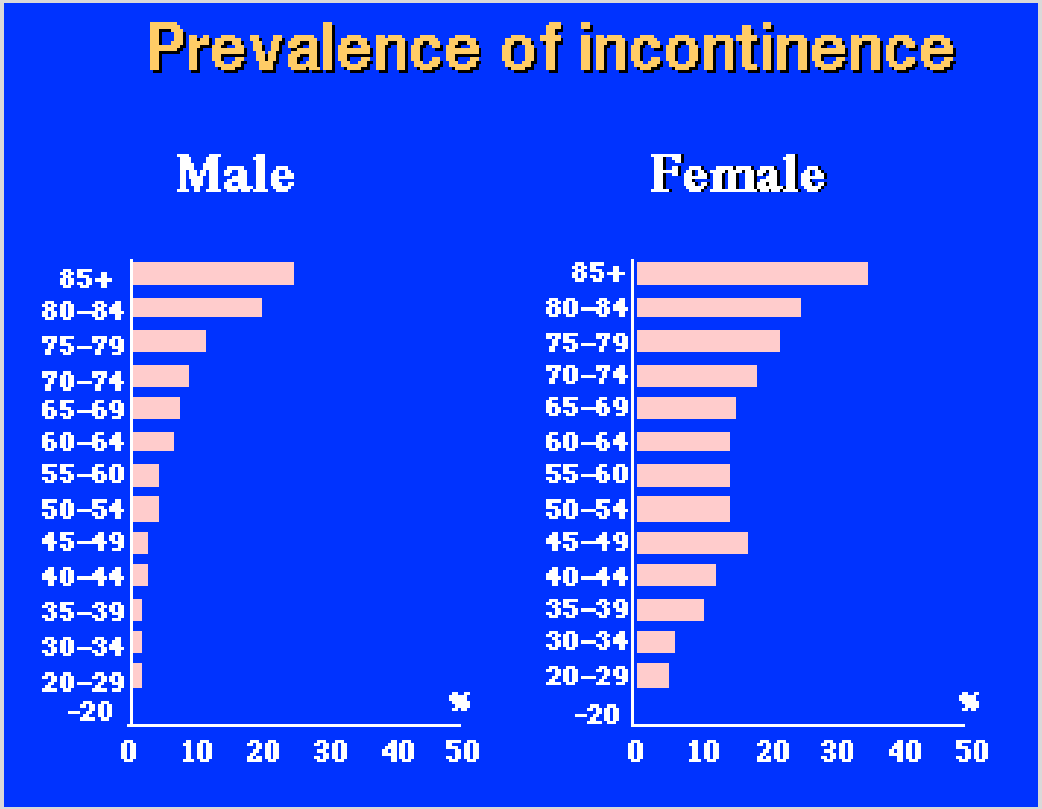


Incontinence

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- Under diagnosed and underreported problem that increases with age
- Affecting 50-84% of the elderly in long-term care facilities
- 2 times more common in females than in males

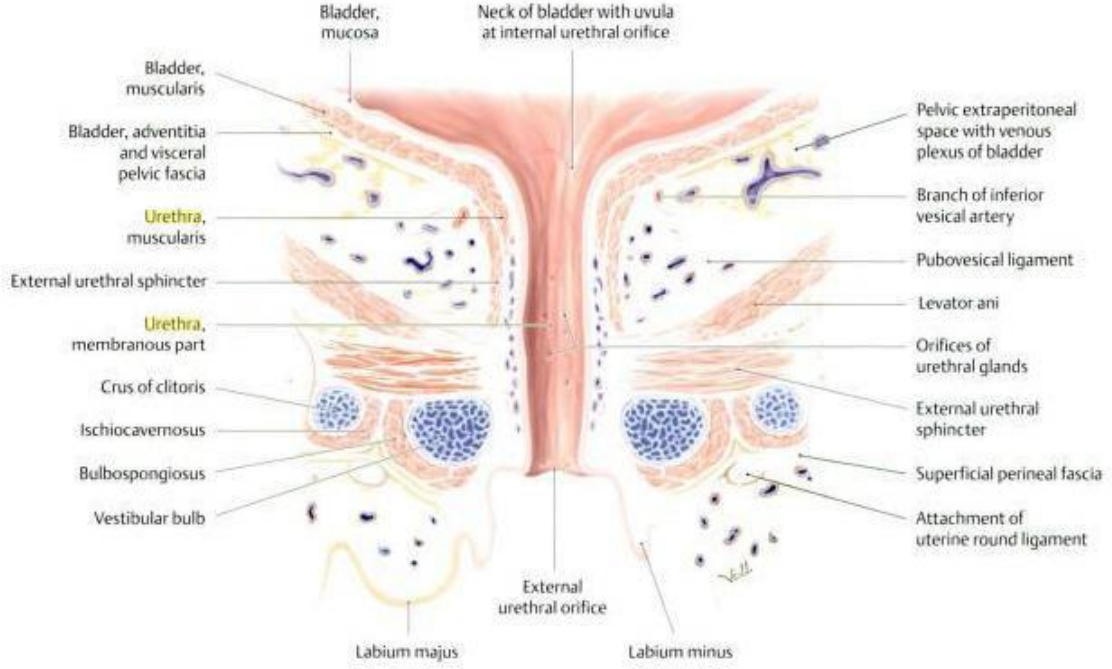
Incontinence



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Incontinence: Pelvic Floor Anatomy



E Female urethra in longitudinal section
Coronal section tilted slightly posterior, anterior view. Unlike the male urethra, the female urethra is straight and only about 3–5 cm long. Thus it is much easier to catheterize than in the male. At the same time, the short length of the female urethra inereases susceptibility to urinary tract infections.

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Incontinence: Diagnosis

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- History
- Physical examination
- Urinalysis
- Voiding diary
- Cotton swab test
- Cough stress test
- Measurement of postvoid residual (PVR) urine volume
- Cystoscopy
- Urodynamic studies

Incontinence: History

- History of Present Illness
- Past medical history
- Past surgical history
- Obstetrical history
- Social history
- Medications

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Incontinence: Medical Conditions

- Chronic cough
- Chronic obstructive pulmonary disease (COPD)
- Congestive heart failure
- Diabetes mellitus
- Obesity
- Connective tissue disorders
- Postmenopausal hypoestrogenism
- CNS or spinal cord disorders
- Chronic UTIs
- Urinary tract stones
- Benign prostatic hyperplasia
- Cancer of pelvic organs

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Incontinence: Medications

- Cholinergic or anticholinergic drugs
- Alpha-blockers
- Over-the-counter allergy medications
- Estrogen replacement
- Beta-mimetics
- Sedatives
- Muscle relaxants
- Diuretics
- Angiotensin-converting enzyme (ACE) inhibitors

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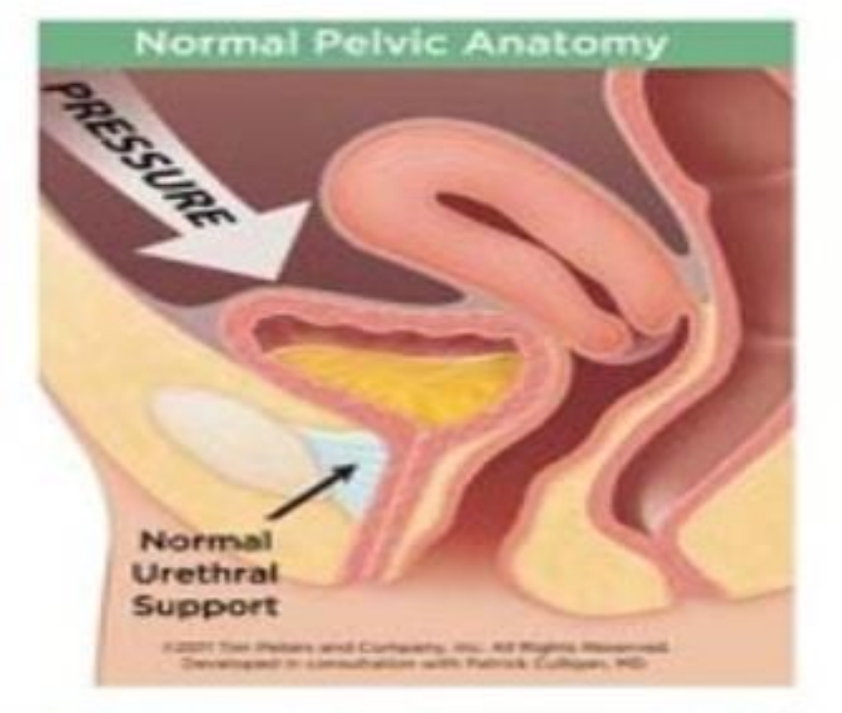
Stress Incontinence: Causes

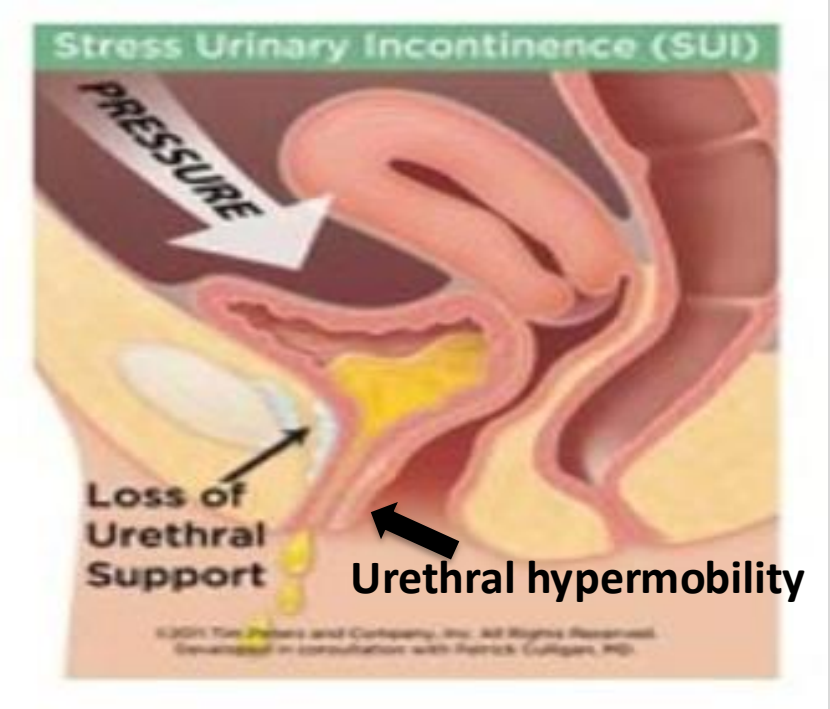
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- Increase in intra-abdominal pressure → raises pressure within the bladder →exceeds urethra's resistance to urinary flow.
  1. Urethral hypermobility due to impaired support from pelvic floor
  2. Intrinsic sphincter deficiency "lead pipe urethra"
     - Loss of function of both the internal and the external sphincter mechanism

Stress Incontinence

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Stress Incontinence: Causes

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- Damage to the nerves, muscle, and connective
- Injury during childbirth
  - Levator ani muscle tears
  - Connective tissue breaks
  - Pudendal/pelvic nerve denervation
- Aging
- Hypoestrogenism
- Chronic connective tissue strain due to primary loss of muscular support
- Activities or medical conditions resulting in long-term repetitive increases in intra-abdominal pressure

Urge Incontinence

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- Urge incontinence is involuntary urine loss associated with a feeling of urgency
- *Detrusor overactivity*→ involuntary detrusor contractions during filling cystometry
- Urge incontinence may be a result of detrusor myopathy, neuropathy, or a combination of both
- When a definable causative neuropathic disorder exists, the coexisting urinary incontinence disorder is termed *neurogenic detrusor overactivity*
- Symptoms of overactive bladder or urge incontinence in the absence of neurologic causes are known as *detrusor instability*

Mixed incontinence

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- Combination of stress and urge incontinence
- Females, 40-60%
- Bladder outlet is weak and the detrusor is overactive
- More common in older patients with urinary incontinence disorders
- Stress incontinence symptoms precede urge incontinence
- Some patients with stress incontinence have urine leakage into the proximal urethra that may, at first, trigger sensory urgency and/or bladder contractions, which initially are suppressible

Male Incontinence

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- BPH may result in urge incontinence
- Androgen-induced enlargement of nodules of glandular tissue
- Increased alpha tone in prostatic and urethral smooth muscle
- Detrusor dysfunction may consist of impaired contractility, detrusor overactivity, or both
- In severe cases of obstruction → retention →overflow incontinence → upper urinary tract damage

Reflex incontinence

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- Due to neurologic impairment of the central nervous system i.e. stroke, Parkinson's disease, brain tumors, spinal cord injury, multiple sclerosis, etc.
  - Suprapontine or suprasacral spinal cord lesions → urge incontinence → detrusor hyperreflexia

Overflow Incontinence

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- Incomplete bladder emptying secondary to impaired detrusor contractility or bladder outlet obstruction
- Impaired detrusor contractility is typically neurogenic in nature; causes include diabetes mellitus, lumbosacral nerve disease from tumors, meningomyelocele, MS, prolapsed intravertebral disks, and high spinal cord injuries
- Less common causes of overflow incontinence include AIDS, genital herpes affecting the perineal area, and neurosyphilis
- The maximal storage capacity of the bladder is reached, oftentimes without the individual realizing that this has occurred
- Incontinence occurs off the top of a chronically over-filled bladder
- Effective emptying is not possible because of a contracting detrusor muscle

Functional incontinence

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- Difficulty reaching the toilet because of physical or psychological impediments
- The etiology of the incontinence may be iatrogenic, environmental, situational, or disease related. The following common mnemonic, DIAPPERS, is helpful in remembering the functional contributors to incontinence :

D - Delirium

I - Infection, urinary

A - Atrophic urethritis or vaginitis

P - Pharmacologic agents

P - Psychiatric illness

E - Endocrine disorders

R - Reduced mobility or dexterity

S - Stool impaction

Nighttime Incontinence

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- Major problems in the older population
- Nighttime bed-wetting may benefit from fluid restriction
- Elimination of caffeine
- DDAVP decreases nighttime urine production → decrease nocturia → hyponatremia especially in elderly patients
- Behavior techniques to help i.e. support hose and/or medications
- Lower extremity edema of the lower extremities
  - Advise elevation of lower extremities several hours during the late afternoon/evening; stimulate natural diuresis, limit the amount of edema present at bedtime
  - Support hose or intermittent sequential compression devices (SCDs) used briefly at the end of the day can reduce lower extremity edema and minimize night time diuresis, thus improving sleep

Continuous Incontinence

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- Vesicovaginal fistula
- Ureterovaginal fistula
  - OB/GYN complications
  - Difficult labor
  - Hysterectomy with enlarged fibroids
  - Increased with emergent surgery
  - Increased with radiation

Cauda equina syndrome

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- S5 nerve root injury (herniation) can cause bladder dysfunction
- Symptoms include :
  - bilateral leg pain and weakness,
  - saddle anesthesia
  - urinary retention or incontinence, and fecal retention or incontinence
  - It is important to recognize this syndrome early because there is a high risk for chronic neurologic deficits if treatment is delayed

Medication Resulting in Incontinence

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- Anticholinergic properties or side effects - urinary retention and thus overflow incontinence
- Alpha-adrenergic agonists - urinary retention and thus overflow urinary incontinence
- Alpha-antagonist - urethral relaxation
- Diuretics – overwhelming of bladder capacity in elderly persons

Medication Resulting in Incontinence

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- Calcium channel blockers - urinary retention with overflow incontinence
- Sedative-hypnotics - immobility secondary to sedation, leading to functional incontinence
- Angiotensin-converting enzyme (ACE) inhibitors – diuretic effect
- Antiparkinsonian medications - urinary urgency and constipation

Urodynamics Study

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- Urodynamics are a means of evaluating the pressure-flow relationship between the bladder and the urethra for the purpose of defining the functional status of the lower urinary tract.
- Multichannel urodynamic studies are more complex than simple urodynamics.

Treatment for Urinary Incontinence

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- **Stress incontinence** - Pelvic floor physiotherapy, anti-incontinence devices, and surgery
- **Urge incontinence** - Changes in diet, behavioral modification, pelvic-floor exercises, and/or medications and new forms of surgical intervention
- **Mixed incontinence** - Pelvic floor physical therapy, anticholinergic drugs, and surgery
- **Overflow incontinence** - Catheterization regimen or diversion
- **Functional incontinence** - Treatment of the underlying cause

Medical treatments

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- In stress and urge urinary incontinence, the following medications may provide some benefit:
  - Alpha-adrenergic agonists
  - Anticholinergic agents
  - Antispasmodic drugs
  - Tricyclic antidepressants
  - Estrogen
  - Alpha-adrenergic blockers
  - Botulinum toxin

Anticholinergics

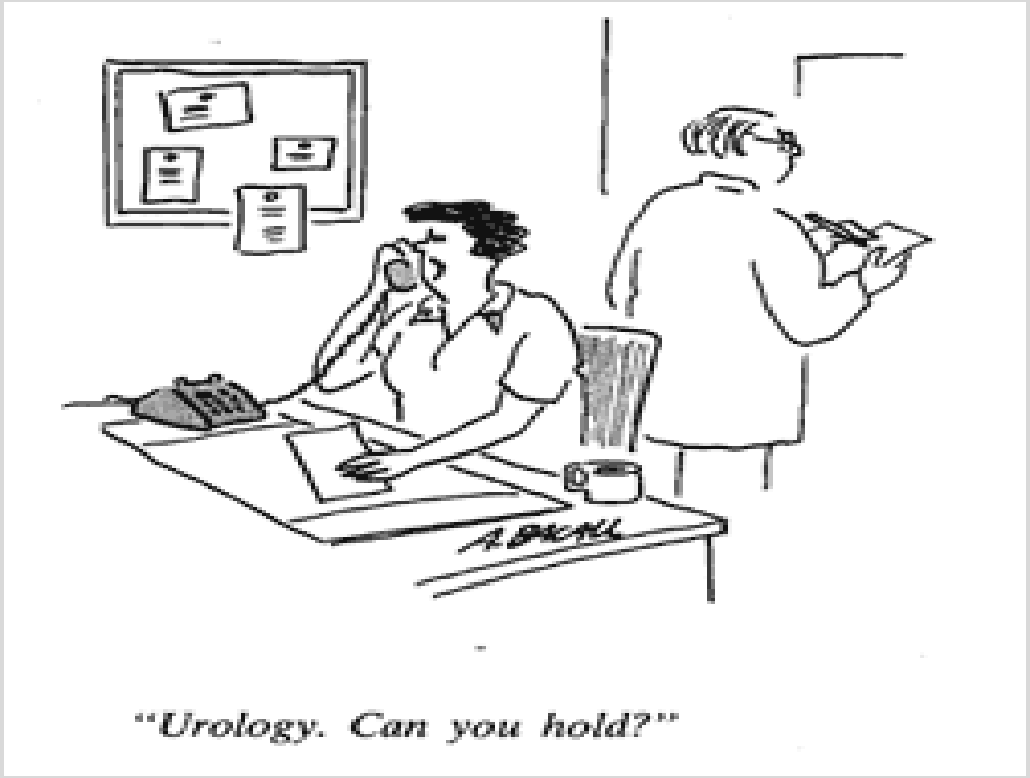
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- Oxybutinin, Tolterodine, Imipramine
- Cause dry mouth, constipation, dry eyes, blurred vision, orthostatic hypotension, and increased heart rate
- Should be avoided by patients with heart disease and closed-angle glaucoma

Surgical Treatments

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- Stress Urinary Incontinence
  - Bladder neck suspension
  - Periurethral bulking therapy
  - Mid-urethral slings
  - Artificial urinary sphincter
- Urge incontinence
  - Sacral nerve modulation
  - Injection of neurotoxins such as botulinum toxin
  - Bladder augmentation



"Urology. Can you hold?"

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Summary Slide

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- Recall micturition versus incontinence
- Define different types of incontinence
- Distinguish causes and treatment for different types of incontinence

Required Resources

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- ScholarRX Bricks
  - Micturition and Incontinence
- Suggested/Supplemental Resources:
  - Urinary Incontinence and Pelvic Prolapse: Epidemiology and Pathophysiology. In: Campbell, Walsh, Wein, eds. *Urology*, *12e*.
  - Elsevier; 2021. Urinary Incontinence. In: Hoffman BL, Schorge JO, Halvorson LM, Hamid CA, Corton MM, Schaffer JI. eds. *Williams Gynecology, 4e*. McGraw Hill; 2020.

HOW TO CONTACT

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Student Hours: Fridays 11a-12p (CT)/ 12p-1p (ET)

- In-Person: Clinical Medicine Department (424D)
- Zoom: https://nyit.zoom.us/my/aloftisdo
- Email questions: aloftis@nyit.edu

Lecture and Lab Feedback Form:

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